



provide mm-level accuracy necessary for industrial automation.

Accerion
www.accerion.tech

Chip in watch for contactless payment

A tiny security chip integrated into a watch enables fast and secured transactions by radio frequency within milliseconds. The chip proves the authenticity with an individual signature and creates a cryptogram of card data, payment



amount, and place of payment. Communication between the card or wearable and the reader is based on Near Field Communication (NFC) technology. From two to ten centimetres away, the chip uses the energy field of the reader to calculate, encrypt, and transmit the data. After successful verification, the bank confirms the payment to the reader.

Infineon Technologies
www.infineon.com

Energy-saving power management ICs

Providing significant energy savings in data centres, the EP70xx series of PMICs integrates triple-output DC-DC power supply into a single 5mm x 5mm

package with no external components while ensuring high current density, accuracy, and fast dynamic voltage. The EP70xx family displays peak efficiencies of up to 91% with a nearly flat efficiency



curve up through 10A of output current and lossless processor state changes that can save 30% or more of processor power. This simplifies the adoption of DC-DC converters by power designers who need multilevel voltage and current regulation.

Empower Semiconductor
www.empowersemi.com

Small-sized automotive LiDAR ICs

LiDAR sensors are becoming increasingly significant in providing accurate distance measurement of objects. These sensors have more than twice the bandwidth and the ability to accommodate 33% more channels within the same module size. Now, automotive makers involved in designing self-driving sys-



tems can enable high-speed autonomous driving with the help of small LiDAR ICs that allow fast autonomous driving at highway speeds by providing more than 2x higher bandwidth and accommodating 32 additional channels within the same module size. These ICs also meet stringent safety requirements of the automotive industry with AEC-Q100 qualification, enhanced electrostatic

discharge performance, and Failure Modes, Effects and Diagnostic Analysis (FMEDA).

Maxim Integrated
www.maximintegrated.com

Low-power ICs for appliances

With increasing demand for smart appliances, there is a need for lower standby power consumption. The BM1ZxxxFJ, an integrated zero-cross detection IC series, solves this issue. A zero-cross detection circuit detects the 0V point of the AC power supply to efficiently control both motors and microcon-



trollers. When the motor stops, pulse control provides greater circuit safety by reliably stopping the voltage at 0V. The BM1ZxxxFJ eliminates any complex design using discrete components and does not use a photo-coupler, thus reducing current consumption and increasing long-term reliability. The IC has been optimised for home appliances such as vacuum cleaners, washing machines, and air-conditioners.

ROHM Semiconductor
www.rohm.com

Stray-field HAL position sensor

The HAL 3970 sensor has highly accurate angular detection from 0 degree to

